Problem 1

(LP Formulations) In lecture, Professor Brito mentioned how computational techniques, like interior-point methods, can solve Linear Programming (LP) problems very efficiently and with a high degree of accuracy. Thus, one could argue the more difficult challenge is formulating a problem as an LP, which is exactly the goal of this problem.

Consider a scenario involving an autonomous drone. The goal is to navigate the drone from its initial position to a desired final position while minimizing the total energy consumption. The vehicle operates in 3D space, so its state $\boldsymbol{x}(t) \in \mathbb{R}^6$ is given by $[x, y, z, \dot{x}, \dot{y}, \dot{z}]$ for $t = 0, \ldots, N$. Furthermore, the drone is controlled by four brushless motors, whose actuator signals are given by $\boldsymbol{u}(t) \in \mathbb{R}^4$, for $t = 0, \ldots, N-1$. The dynamics of the system is given by the linear relation

$$\boldsymbol{x}(t+1) = \boldsymbol{A}\boldsymbol{x}(t) + \boldsymbol{B}\boldsymbol{u}(t), \quad t = 0, \ldots, N-1, \text{“} \tag{1}$$

where $\boldsymbol{A} \in \mathbb{R}^{6\times 6}$ and $\boldsymbol{B} \in \mathbb{R}^{6\times 4}$ are given. We assume that the initial state is zero, i.e., $\boldsymbol{x}(0) = \boldsymbol{0}$. Our goal is to choose the inputs $\boldsymbol{u}(0), \ldots, \boldsymbol{u}(N-1)$ to minimize the energy use, which is given by

$$E = \sum_{t=0}^{N-1} \sum_{i=1}^{4} f(u_i(t)), \tag{2}$$

subject to the constraint that $\boldsymbol{x}(N) = \boldsymbol{x}_{\text{stop}}$, where $\boldsymbol{x}_{\text{stop}} \in \mathbb{R}^6$ is a (given) desired target state. The function $f \colon \mathbb{R} \to \mathbb{R}$ is the energy consumption map for an actuator. It represents the amount of energy consumed as a function of the actuator signal amplitude. In this problem we use

$$f(a) = \begin{cases} |a| & \text{if } |a| \le 1, \\ 3|a| - 2 & \text{if } |a| > 1. \end{cases} \tag{3}$$

In other words, we set energy use to be proportional to the absolute value of the actuator signal (for small actuator signals; for larger actuator signals the efficiency is reduced by a factor of three.) Your task is as follows: **Formulate this optimal control problem as an LP in standard form, as described in class.** We expect: (1) An explanation regarding the formulation of each of the constraints, (2) A clear description of the formulated LP, and (3) A conversion of the LP to standard form.

Note: Try to keep your answer reasonably concise, with appropriate vector representations of the data.

Problem 1 Solution

There are two key insights

1. We begin by trying to “solve” the dynamical system, which in essence is a recurrence. Of course, we cannot, but just stick with us. One way to solve

a recurrence is try to express bigger values in terms of smaller ones. In this case, we have

$$
\begin{aligned}
\boldsymbol{x}(1) &= \boldsymbol{A}\boldsymbol{x}(0) + B\boldsymbol{u}(0) = B\boldsymbol{u}(0) \\
\boldsymbol{x}(2) &= \boldsymbol{A}B\boldsymbol{u}(0) + B\boldsymbol{u}(1) \\
x(N) &= \boldsymbol{A}^{N-1}\boldsymbol{B}\boldsymbol{u}(0) + \boldsymbol{A}^{N-2}\boldsymbol{B}\boldsymbol{u}(1) + \cdots + \boldsymbol{A}\boldsymbol{B}\boldsymbol{u}(N-1) + \boldsymbol{B}\boldsymbol{u}(N-1) = \boldsymbol{x}_{final}
\end{aligned}
$$

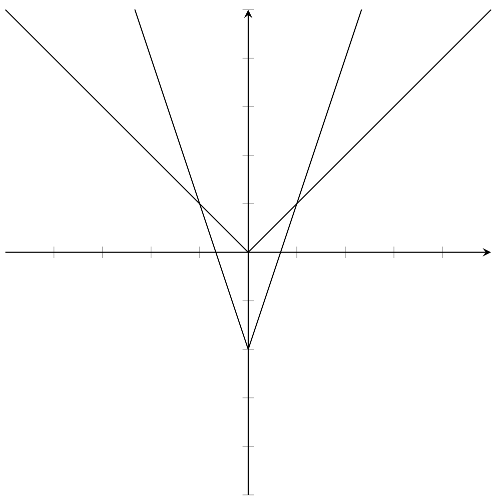
We now define the controllability matrix:

$$C = \begin{bmatrix} \boldsymbol{A}^{N-1}\boldsymbol{B} & \boldsymbol{A}^{N-2}\boldsymbol{B} & \cdots & \boldsymbol{A}\boldsymbol{B} & \boldsymbol{B} \end{bmatrix}$$

Which we can substitute to our equation to get our "navigation" constraint:

$$\boldsymbol{C}\boldsymbol{u} = \boldsymbol{x}$$

ofc this only holds for one of the u_i, so **you have to do this four times for each of the u_i**



2. Now we must turn f into a linear objective. Instead of minimizing f, we minimize some variable t, where

$$
\begin{aligned}
|a| &\leq t \\
3|a| - 2 &\leq t
\end{aligned}
$$

To see that this works, note that for $|a| \leq 1$, we have $|a| \geq 3|a| - 2$, while for $|a| > 1$, $|a| \leq 3|a| - 2$. Then for $|a| \leq 1$, the second condition is redundant, while for $|a| > 1$, the first is redundant, which gives us exactly what we want. Again, this must be repeated for each of the time steps and each of the f_i.

Everything else is just matrix and vector manipulations. The final answer is synatatically correct if:

a. The objective function can be expressed as $c^T x$ for some **vector** c (NOT A MATRIX)

b. The constraints are all of the form $\boldsymbol{Ax} \leq \boldsymbol{b}$ (yes, you should take off points if inequalities are facing the wrong direction, the direction of the inequality was explicitly defined in lecture).

c. All the decision variables are nonnegative.

Our final answer is

$$\underset{\boldsymbol{u}}{\text{minimize}}\ \mathbf{1}t$$

$$subjectto$$

$$\boldsymbol{Cu} = \boldsymbol{x}$$

Problem 2 Solution

a) The absence of arbitrage condition states that there exists no vector $\boldsymbol{x}$ such that $\boldsymbol{x}^T\boldsymbol{R}^T \geq \mathbf{0}^T$ and $\boldsymbol{x}^T\boldsymbol{p} < 0$. This is of the same form as condition (b) in the Statement of Farkas' lemma. (note that here $\boldsymbol{p}$ plays the role of $\boldsymbol{b}$, and $\boldsymbol{R}^T$ plays the role of $\boldsymbol{A}$. Therefore, by Farkas' lemma, the absence of arbitrage condition holds if and only if there exists some nonnegative vector $\boldsymbol{q}$ such that $\boldsymbol{R}^T\boldsymbol{q} = \boldsymbol{p}$, which is the same as the condition in the theorem's statement.

b) We construct the payoff matrix:

$$R = \begin{pmatrix} uS & r & \max(0, uS-K) \\ dS & r & \max(0, dS-K) \end{pmatrix}$$

The equivalence proved in part (a) along with the fact that the absence of arbitrage condition holds allows us to conclude that there exists a q where $q \geq 0$ such that $p = q^T R$. Let $q = [\gamma, \delta]$. Then, expanding the product:

$$p = \begin{pmatrix} \gamma uS + \delta dS \\ \gamma r + \delta r \\ \gamma \max(0, uS-K) + \delta \max(0, dS-K) \end{pmatrix} = \begin{pmatrix} S(\gamma u + \delta d) \\ r(\gamma + \delta) \\ \gamma \max(0, uS-K) + \delta \max(0, dS-K) \end{pmatrix}$$

The initial value of the stock is S, so from the first row:

$$S(\gamma u + \delta d) = S$$

$$\gamma u + \delta d = 1$$

The initial value of the dollar invested in the risk-free asset is 1, so from the second row:

$$r(\gamma + \delta) = 1$$

$$\gamma + \delta = \frac{1}{r}$$

And, finally, the initial value of the option is:

$$\gamma \max\{uS - K, 0\} + \delta \max\{dS - K, 0\}$$

Appendix

Q1 is adapted from Boyd and Vanderberge Q2 is adapted from Introduction to Linear Optimization by Bertimas